Numerical Analysis (089180) A.Y. 2024-2025
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Lab 4 – Homework Solutions

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A Homework

Homework 4.1. Consider the linear system $\mathbb{E}\boldsymbol{x} = \boldsymbol{b}$ with

$$\mathbb{E} = \begin{bmatrix} 4 & 1 & 1 & 1 & 5 \\ 4 & 1 & 2 & 0 & 0 \\ 1 & 0 & 15 & 5 & 1 \\ 0 & 2 & 4 & 10 & 2 \\ 3 & 1 & 2 & 4 & 20 \end{bmatrix} \qquad \text{and} \qquad \boldsymbol{b} = \begin{bmatrix} 12 \\ 19 \\ 22 \\ 18 \\ 30 \end{bmatrix}$$

1. Verify with Matlab that the sufficient conditions for the existence and uniqueness of the LU decomposition without pivoting are not fulfilled by matrix $\mathbb{E}$.
2. Compute the LU of matrix $\mathbb{E}$ with the Matlab command `lu` and verify if pivoting takes place with the Matlab command `spy`.
3. Solve the system $\mathbb{E}\mathbf{x} = \mathbf{b}$ by using the LU decomposition at the previous item.

Solution Homework 4.1.

hw_4_1.m

```
%% Homework 4.2
clc
clear all
close all

E = [4 1 1 1 5;
    4 1 2 0 0;
    1 0 15 5 1;
    0 2 4 10 2;
    3 1 2 4 20];

b = [12 19 22 18 30]';

n = size(E,1);
%% Verify with Matlab that the sufficient conditions for the existence and
% uniqueness of the LU decomposition without pivoting are not fulfilled
% by matrix E

% Let calculate the determinant of the all the minor fo the matrix E

for i = 1:n-1
   det(E(1:i, 1:i))
end

% Since one of the minor determinants is equal to zero, the LU
% factorization without pivoting does not exist!
```

```
%% Compute the LU of matrix E with the Matlab command lu and verify
% if pivoting takes place with the Matlab command spy.
%%%%%%%%%%%%%%%%%%%%%%%%%%%%% ATTENTION%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
 % if a matrix E does not admit a LU factorization and you
 % call lu command without the output P, matlab does not
 % return the lu factorization of P*E but a factorization
 % of E which is not lower-upper triangular
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

[L, U] = lu(E);

L % is not triangular!
figure()
% spy plots the sparsity pattern of matrix L.
% Nonzero values are colored while zero values are white
spy(L)
U

[L, U, P] = lu(E);

L
figure()
spy(L)
U
P % is not the identity matrix => pivoting is needed

%% Solve the system Ex = b by using the LU decomposition at the
% previous item.
% backward substitutions (you can use also backward_substitution.m)
y = L\(P*b);
% forward substitutions (you can use also forward_substitution.m)
x = U\y
xex = E\b
```

Homework 4.2. Consider the system $\mathbb{A}\boldsymbol{x} = \boldsymbol{b}$ with

$$\mathbb{A} = \begin{bmatrix} 1 & 2 & 1 & 1 \\ 1 & 4 & 0 & 2 \\ 2 & 10 & 4 & 0 \\ 1 & 0 & 2 & 2 \end{bmatrix} \qquad \text{and} \qquad \boldsymbol{b} = \begin{bmatrix} 3 \\ 3 \\ 10 \\ 1 \end{bmatrix}.$$

a. Compute the solution of the system $\mathbb{A}\boldsymbol{x} = \boldsymbol{b}$ using the $\mathbb{LU}$ decomposition with pivoting of matrix $\mathbb{A}$.

b. Find the determinant of $\mathbb{A}$ (use the Matlab command `det` only to compute the determinant of $\mathbb{P}$).

Solution Homework 4.2.

hw_4_2.m

```
%% Homework 4.2
clc
clear all
close all
%% Compute the solution of the system Ax = b using the LU decomposition
% with pivoting of matrix A.
A = [1 2 1 1;
    1 4 0 2;
    2 10 4 0;
    1 0 2 2];

b = [3 3 10 1]';

[L, U] = lu(A);
figure()
spy(L) % L is not lower triangular => pivoting is needed
```

```
[L, U, P] = lu(A);
figure()
spy(L) % OK!

y = L\(P*b);
x = U\y

xex = A\b

%% Find the determinant of A (use the Matlab command det only to compute
%the determinant of P).
detA = prod(diag(U))/det(P)
det(A)

% Thanks to the usual Binet formula, we have det(PA) = det(P)det(A) =
% det(LU) = det(L)det(U)
```

Homework 4.3. Consider the tridiagonal matrix $\mathbb{A}_n \in \mathbb{R}^{n\times n}$ defined as

$$\mathbb{A}_n = \begin{bmatrix} a & b & & & \\ b & a & b & & \\ & \ddots & \ddots & \ddots & \\ & & b & a & b \\ & & & b & a \end{bmatrix}$$

for $a = 2$ and $b = 1$.

a. Verify with MATLAB that $\mathbb{A}_{10}$ is a symmetric positive definite matrix.

b. Provide the form of the matrix $\mathbb{V}_{10}$ such that $\mathbb{A}_{10} = \mathbb{V}_{10}^T \mathbb{V}_{10}$

Solution Homework 4.3.

hw_4_3.m

```
%% Homework 4.5
clc
clear all
close all

a = 2;
b = 1;
n = 10;
A = diag(a * ones(n, 1)) + diag(b * ones(n-1,1), -1) + diag(b * ones(n-1,1), +1);
%% Verify with MATLAB that A_10 is a symmetric positive definite matrix

eig(A)
% All the eigenvalues are positive and it is trivial to prove the simmetry.

%% Provide the form of the matrix V10 such that A10 = VT10 V10
V = chol(A)
format short e
V'*V - A
% V is bidiagonal since Cholesky decomposition preserves the pattern of the
% matrix.
```